

Analyzing Bias in Cultural Content: A Word Embedding Approach to Film Genre Classification

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Abstract

Natural Language Processing has witnessed rapid advancements, with word embeddings playing a crucial role in vectorizing text data and learning semantic similarities. However, these embedding techniques can reflect biases present in the training data, potentially leading to societal issues. Analyzing bias in cultural content is thus an important research topic. This study measures biases across film genres to verify whether specific genres exhibit distinct linguistic characteristics. We apply the Word Embedding Association Test (WEAT) to film plot data to determine if certain genres are more strongly associated with particular sets of words. The results will contribute to understanding linguistic differences between film genres and assessing the validity of existing genre classifications, benefiting both the film industry and bias research in cultural content.

1 Introduction

Natural Language Processing (NLP) has experienced significant advancements recently, with word embeddings playing a vital role in vectorizing text data and learning semantic similarities. These techniques enable machines to understand the relationships between words and concepts, facilitating various applications such as machine translation, sentiment analysis, and information retrieval.

However, word embeddings are susceptible to reflecting biases present in the training data. These biases can perpetuate stereotypes and discriminatory patterns, leading to unfair or harmful outcomes in downstream applications. For example, if a word embedding model is trained on a corpus containing gender stereotypes, it may associate certain professions more strongly with one gender than the other.

Analyzing bias in cultural content is particularly important due to its potential impact on shaping public perceptions and attitudes. Films, as a powerful medium of storytelling, can reinforce or challenge societal norms and values. Therefore, it is crucial to investigate whether specific film genres exhibit linguistic biases that may contribute to societal issues.

In this study, we focus on measuring biases across film genres to verify whether specific genres exhibit distinct linguistic characteristics. We apply the Word Embedding Association Test (WEAT) to film plot data to determine if certain genres are more strongly associated with particular sets of words. By comparing the linguistic profiles of different film genres, we aim to uncover potential biases and gain insights into the underlying cultural representations.

Our research contributes to the growing body of work on bias detection and mitigation in NLP. By specifically focusing on film genres, we provide a nuanced understanding of how linguistic biases manifest in cultural content. The results of this study can inform filmmakers, content creators, and policymakers about the potential biases in their work and encourage them to promote more inclusive and equitable representations.

2 Related Work

2.1 Word Embeddings and Bias

Word embeddings, such as Word2Vec and GloVe, have become fundamental tools in NLP for representing words as dense vectors in a high-dimensional space. These embeddings capture semantic relationships between words, allowing machines to understand the meaning and context of text. However, it has been shown that word embeddings can inherit biases present in the training data.

Bolukbasi et al. demonstrated that word embeddings trained on large text corpora can exhibit gender stereotypes. For example, they found that the word "computer programmer" was more closely associated with male pronouns than female pronouns. Caliskan et al. further showed that word embeddings can reflect biases related to race, religion, and other sensitive attributes.

Several techniques have been proposed to mitigate biases in word embeddings. Bolukbasi et al. introduced a debiasing algorithm that modifies the embedding space to remove gender stereotypes. Other approaches involve re-weighting the training data or using adversarial training to reduce bias.

2.2 Bias Analysis in Cultural Content

Bias analysis in cultural content has gained increasing attention in recent years. Researchers have investigated biases in various forms of media, including literature, news articles, and social media.

Sap et al. analyzed the presence of stereotypes in natural language interactions and found that certain demographic groups are more likely to be associated with negative stereotypes. Recasens et al. developed a linguistic model to detect biased language in news articles.

Several studies have focused on bias analysis in films. A number of researchers have examined gender representation in films, analyzing the portrayal of female characters and their roles in the narrative. Other researchers have investigated racial and ethnic stereotypes in films, examining the representation of different racial groups and their associated characteristics.

3 Methodology

3.1 Data Collection and Preprocessing

We collected movie plot synopses spanning from 2001 to August 2019 from an online movie database. The dataset was divided into two categories: art films and mainstream (commercial) films, stored separately as `synopsis_art.txt` and `synopsis_gen.txt`.

The text data was preprocessed using the following steps:

1. **Morphological Analysis:** We used the Okt morphological analyzer to split the text into morphemes and extract nouns. Noun extraction is beneficial for identifying the main concepts of the movies and is suitable for analyzing semantic relationships rather than emotional elements.
2. **Stop Word Removal:** We removed stop words (e.g., ,) that have little meaning to improve the accuracy of the analysis.
3. **TF-IDF Calculation:** We calculated the Term Frequency-Inverse Document Frequency (TF-IDF) to select representative words for each movie genre. TF-IDF measures the importance of a word within a document, allowing us to extract words that best represent the characteristics of each genre.
4. **Word2Vec Model Training:** We trained a Word2Vec model on the preprocessed text data to generate word embeddings. Word2Vec learns semantic similarities between words in a vector space, which is used to analyze the association of specific movie genres with certain words.

3.2 Word Embedding Association Test (WEAT)

WEAT is a method for quantifying the association between two sets of concepts (X, Y) and two sets of attributes (A, B) based on their similarities. The WEAT score is defined as follows:

$$s(X, Y, A, B) = \frac{\text{mean}_{x \in X} s(x, A, B) - \text{mean}_{y \in Y} s(y, A, B)}{\sigma_{X, Y}}$$

Here, $s(w, A, B)$ represents the association of a specific word w with attribute sets A and B , and $\sigma_{X, Y}$ normalizes the variance between concept sets X and Y .

In this study, set X represents words associated with commercial films, and set Y represents words associated with art films. The attribute sets (A, B) represent commerciality and artistry, respectively (e.g., 'blockbuster', 'popular' vs. 'experimental', 'auteur'). WEAT analysis was performed to evaluate how strongly a specific movie genre is associated with a particular attribute word.

3.3 TF-IDF-Based Representative Word Extraction

We extracted and compared words representing art films and mainstream films using TF-IDF. This process allowed us to extract the top 100 words representing each genre. Additionally, we analyzed words that uniquely represent each genre without overlapping.

4 Experiments and Results

4.1 WEAT Score Calculation and Heatmap Analysis

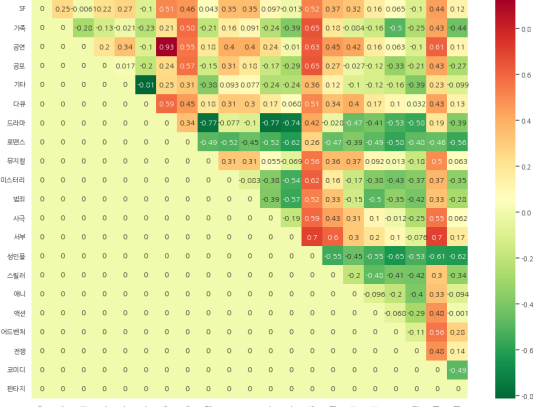


Figure 1: Heatmap with $\text{min_df} = 0.85$

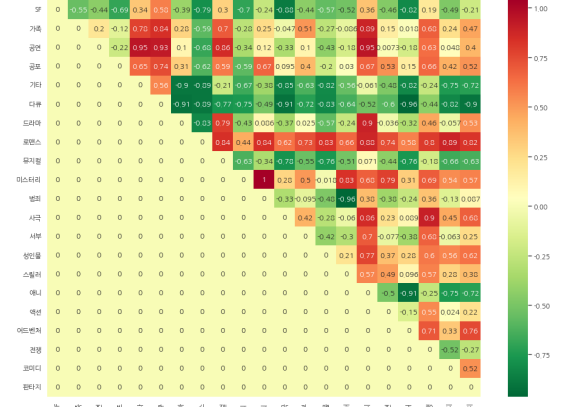


Figure 2: Heatmap without min_df

The WEAT analysis results revealed associations between movie genres. High similarities were observed between drama and romance, and between horror and thriller genres, which can be attributed to the complex narrative structures and the mixing of genre elements in movies. Given the trend of hybrid genre movies in the film market, WEAT analysis provides objective evidence supporting this trend.

The heatmap visualization, comparing genre similarities, showed the impact of the min_df parameter on genre correlations. As shown in Figures 1 and 2, without setting min_df (including common words frequently appearing in all documents), the overall correlation was high. However, setting min_df to 0.85 (removing rare words appearing only in specific genres) reduced the overall genre correlation. This indicates that setting min_df clarifies the differences between genres by removing rare words. Specifically, genres like family, performance, and documentary showed a significant decrease in correlation, suggesting they use more unique vocabulary compared to other genres.

In detail, without min_df , high correlations were noticeable between drama-romance and horror-thriller. After setting min_df , these correlations weakened, suggesting that only general characteristics between genres remained after removing rare words. Conversely, the slight increase in correlation between comedy and fantasy after setting min_df suggests that certain common words in these two genres became more prominent after rare word removal.

4.2 TF-IDF-Based Representative Word Analysis

The analysis of representative words extracted based on TF-IDF confirmed distinct lexical differences between art films and mainstream films. Art films frequently featured unique names or specific concepts such as “”, “”, “”, while mainstream films often included popular terms such as “”, “”, “”. This suggests that art films attempt in-depth explorations of specific themes or individuals, while mainstream films are produced for a broader audience. These lexical differences highlight the existence of representative words that reflect the characteristics of each genre, which can contribute to establishing objective criteria for movie genre classification.

4.3 Future Research Directions: In-Depth Analysis Using Pre-trained Models

In this study, we analyzed the linguistic biases between film genres based on our own constructed dataset of movie plot synopses. Future research could enhance the objectivity and generalizability of the analysis by actively utilizing pre-trained Word2Vec models such as the GoogleNews embedding model (`GoogleNews-vectors-negative300.bin.gz`). Because the GoogleNews embedding model is trained on vast amounts of news data, it is advantageous for capturing relationships between various words and concepts, which would allow for a more objective assessment of biases appearing in movie plot data.

The following additional experiments could be performed using pre-trained models:

1. **Sentiment Analysis of Genre-Related Words:** Analyze the tendency of specific words related to movie genres to be associated with positive or negative sentiments more precisely. For example, quantitatively assess whether words related to the comedy genre show higher associations with words expressing positive sentiments, or whether words related to the horror genre show higher associations with words expressing negative sentiments. This analysis could provide a deeper understanding of how movie genres are related to linguistic expressions that evoke specific emotions.
2. **Social Bias Detection in Movie Content:** Utilize pre-trained models to identify social biases in movie content. For example, if words describing specific genders or races are associated with certain genres, it may suggest that the genre is producing content in a way that reinforces social biases. To detect such social biases, words related to gender, race, and occupation could be set as attribute sets, and WEAT scores could be calculated for each movie genre to quantitatively assess bias.
3. **Genre Classification Accuracy Improvement:** Calculate WEAT scores based on extracted words to quantitatively analyze the associations between genres and improve the accuracy of movie genre classification. Since current movie genre classifications are often based on subjective criteria, establishing objective criteria is important. WEAT scores calculated based on words extracted using pre-trained models could be compared with existing genre classifications to verify the validity of the genre classification.
4. **Integration of Diverse Data Sources:** Although this study only used movie plot data for analysis, future research could improve the accuracy of the analysis by integrating diverse data such as movie reviews, ratings, and audience numbers. For example, calculating WEAT scores using words extracted from movie review data together with words extracted from movie plot data could reveal how audience perceptions of movies differ by genre.

These additional experiments are expected to enable in-depth analysis of the linguistic characteristics and social biases of movie content, which can contribute not only to the film industry but also to the development of cultural content in general. Furthermore, the methodology presented in this study can be applied to the bias analysis of other cultural content (e.g., dramas, music, games).

5 Conclusion

In this study, we analyzed biases across film genres using Word2Vec-based WEAT. Focusing on commercial and art films, we quantitatively evaluated how strongly each genre is associated with specific words. The results showed that it is difficult to clearly distinguish between commercial and art films, suggesting that film genres are not simply divided dichotomously but have various mixed attributes and elements.

WEAT scores confirmed high similarities between some genres such as drama and romance, and horror and thriller, which can be interpreted as a phenomenon arising from the fact that film as a medium encompasses complex narrative structures and diverse genre elements. For example, romance elements are often strongly revealed within the drama genre, and the horror genre often utilizes production techniques that evoke thriller-like tension. These results show that the boundaries between film genres are not clear and that various genre elements are mixed in the filmmaking process to reflect the diverse tastes and expectations of the audience.

In addition, comparing the results with and without `min_df` setting through heatmap visualization, we confirmed a tendency for the correlation between genres to decrease overall when rare words were removed (`min_df=0.85`). This can be interpreted as the unique characteristics of each genre being more

emphasized by removing rare words. In particular, the decrease in correlation was prominent in certain genres such as family, performance, and documentary, which suggests that these genres use more unique vocabulary compared to other genres. On the other hand, the slight increase in correlation between comedy and fantasy after setting `min_df` is presumed to be because certain words commonly used in both genres were relatively emphasized.

The TF-IDF-based representative word analysis also revealed distinct lexical differences between art films and mainstream films. Art films frequently featured proper nouns or specific concepts such as “”, “”, “”, reflecting the tendency of art films to attempt in-depth explorations of specific individuals or events. On the other hand, mainstream films mainly included words with popular recognition such as “”, “”, “”, which shows the characteristic that mainstream films are produced for a broader audience.

These results are closely related to the changing trends in the film market. Recently, hybrid films that mix several genre elements without being limited to specific genres are gaining popularity in the film market. This trend can be interpreted as an attempt to satisfy the diverse tastes and needs of the audience, and the WEAT analysis results of this study can be used as evidence reflecting this trend.

This study has provided important insights into the film industry and cultural content research by quantitatively analyzing the linguistic characteristics and biases of film content. However, this study has limitations in that it focused only on text data (plot synopses). Future research needs to derive more comprehensive results by integrating data such as movie reviews, ratings, and audience numbers, in addition to plot synopses. It will also be an important task to enhance the objectivity and generalization of the analysis by utilizing external data such as pre-trained Word2Vec models (GoogleNews-vectors-negative300.bin.gz).

In conclusion, this study has shown that WEAT and TF-IDF-based analysis are effective in identifying biases and linguistic characteristics between film genres. This methodology is expected to be applicable not only to film but also to the bias analysis of other cultural content such as dramas, music, and games. Through in-depth research considering various attributes and data, we will be able to contribute to understanding the linguistic characteristics and social influence of cultural content.

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